

Lab 01

Software Verification and Validation Lab

SWE 4806



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Lab Title

Modeling Systems using Alloy

Objective

The objective of this lab is to introduce students to relational modelling and formal verification using Alloy. Students will learn how to:

- Model real-world systems using Alloy signatures and relations
- Define constraints using facts
- Write predicates, functions, and assertions
- Use Alloy Analyser to detect counterexamples
- Understand how local constraints may still allow unexpected global behaviour

Topics Required Before This Lab

1. Signatures (sig)

A signature defines a set of objects.

```
1 sig Person {}
```

This defines a set called **Person**.

2. Relations

Relations represent associations between objects.

```
1 sig Person {
2   father: lone Man
3 }
```

This means every person may have at most one father.

3. Multiplicity Keywords

Keyword	Meaning
one	Exactly one
lone	Zero or one
some	One or more
set	Any number



4. Facts

Facts define constraints that are always true.

```
1 fact {  
2   no p: Person | p in p.father  
3 }
```

This states that nobody can be their own father.

5. Predicates

Predicates define reusable logical conditions.

```
1 pred example[p: Person] {  
2   some p.father  
3 }
```

6. Functions

Functions return values or sets.

```
1 fun parents[p: Person]: set Person {  
2   p.father + p.mother  
3 }
```

7. Assertions

Assertions are properties we want to verify.

```
1 assert noSelfParent {  
2   no p: Person | p.father = p  
3 }
```

Assertions are checked using the `check` command.

8. Run and Check

Run is used to generate example instances.

```
1 run example for 5
```

Check is used to verify assertions.

```
1 check noSelfParent for 5
```

Relational Operators in Alloy

Union (+)

Combines relations.

```
1 father + mother
```



Transpose ($\sim$)

Reverses a relation.

```
1 wife = ~husband
```

Transitive Closure ($\wedge$)

Used to traverse recursive relationships.

```
1 p.^father
```

This represents all ancestors reachable through the father relation.

System Description

In this lab, students will model a hospital management system. The system contains different types of people responsible for treatment and healthcare services, along with individuals receiving medical services. The system also maintains appointment schedules, ward assignments, and referral relationships among healthcare providers.

Each healthcare provider can manage multiple appointments, while each individual receiving treatment may have appointments with different providers. A patient can be assigned to at most one ward at a time. Healthcare providers may refer patients to other providers for specialized treatment.

The system must satisfy several logical constraints:

- A patient cannot have multiple appointments with the same healthcare provider at the same time.
- Every patient must be assigned to at most one ward.
- A healthcare provider cannot refer a patient to themselves.
- Referral relationships must not create cycles.
- Appointment and ward assignment relationships must remain consistent from both directions.

Naming Convention

Name your files: id_lab02.asl